Virtual reality

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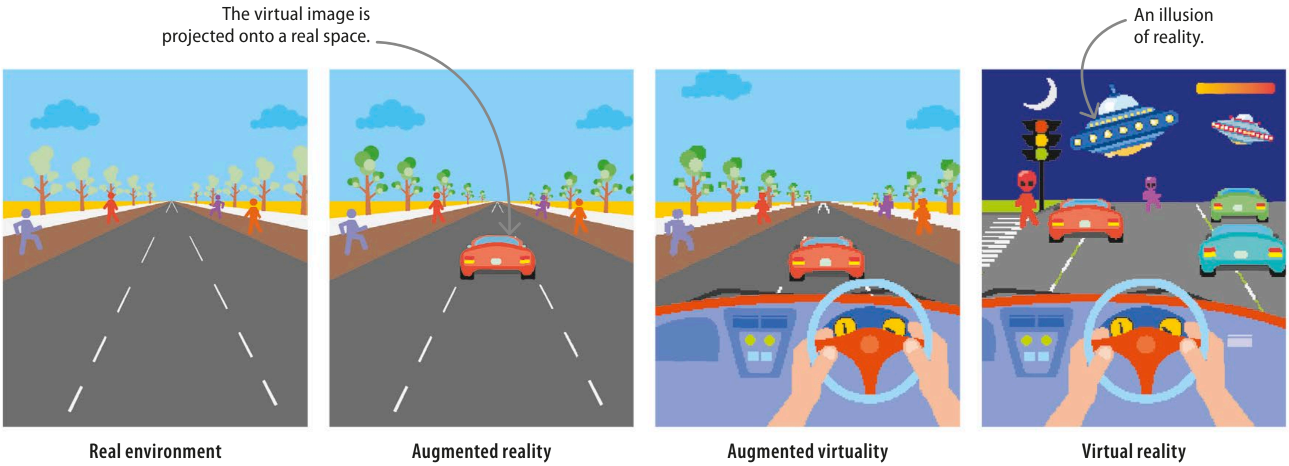
Virtual reality (VR) sounds like something that belongs in science fiction movies and television shows, but the idea has been around for many decades.

Augmented reality vs virtual reality

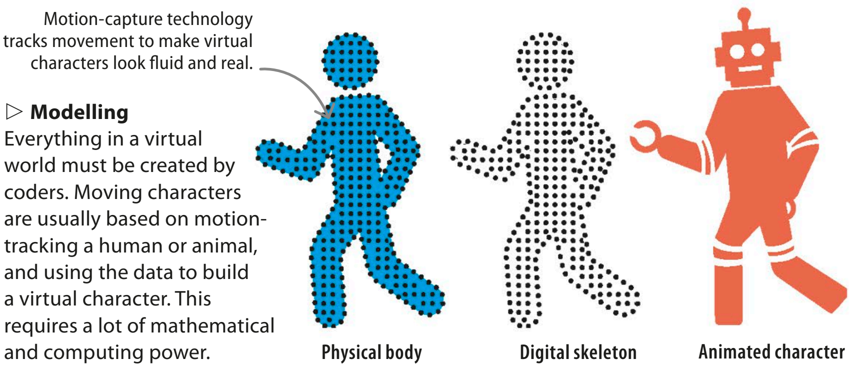
Augmented reality (AR) is a virtual layer over real, physical things. One example is a social media image filter that adds whiskers or dog ears to a human face. Virtual reality, by contrast, creates an entirely new illusion of reality. It includes detailed sights and sounds that make the experience convincing.

▽ **Reality-virtuality spectrum**
Virtual reality can be seen as being the end point in a spectrum that starts as a real environment. Augmented reality adds specific virtual aspects, and augmented virtuality mixes interactive real-world aspects with a mostly virtual world.

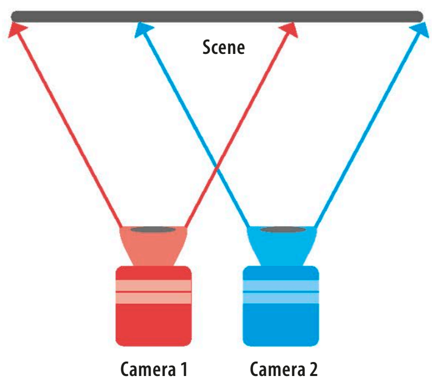


How VR works

For a VR illusion to work, the user needs to feel like they're truly interacting with their environment. VR technology must respond to a user's actions in real time. Any glitch, lag, or gap breaks the illusion. VR requires powerful hardware and software that can predict a user's movements and pre-render images.



▷ **Modelling**
Everything in a virtual world must be created by coders. Moving characters are usually based on motion-tracking a human or animal, and using the data to build a virtual character. This requires a lot of mathematical and computing power.



△ **Depth**
Since our eyes are several centimetres apart, each one captures a different view of the world. This must be recreated by VR for the illusion to work. Using 2D images to create 3D is called stereoscopic vision.